Supplementary Material

Where the Bits Must Live: A Tight Input-Side Information Bound for Fixed-State Spectral Models

Anonymous submission. This document contains the appendices to the main paper. It is self-contained: each proof restates the final premises and the identifier of the claim it establishes, and proofs cross-reference those identifiers rather than referring to 'the proof above'. Numerical witnesses certify non-vacuity and agreement; they are never used as premises.

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The main paper states four core results. SE1-SE5 in Appendix A.7-A.11 are supporting statements about an exactly-defined finite consolidation process; P1-P8 and L1-L3 are propositions and lemmas. No theorem is introduced in the supplementary material.

Appendix A Complete Analytic Proofs

All proofs are analytic. Numerical witnesses certify non-vacuity and agreement with the statements; they are not used as premises.

A.0 Notation and unit ledger

symbol	meaning	unit / where used
B	total input-side bit budget, B = sum_t b_t	bits; Theorem A
b_t	per-position alphabet excess log2|pi^(-1)(v_t)|	bits; Section 2
b	bits per insight per consolidation round	bits; CT7(b), SE4; here b = 5 = log2 V_K
w, T	insights per round, number of rounds	count; CT7(b), SE4
state bytes	structural size of the recurrent state	BYTES, not bits; Table 8, Appendix D
32768	d x d fp64 core state, d = 64 (fusion config)	bytes; Tables 16, 18, 34; DIFFERENT from Table 8
c, o	orthogonal successor rank per hop, tensor order	count; P5, Result 2

Table A0. Uppercase B and lowercase b are different quantities; state bytes is a size in bytes and never a bit budget. Table 8 (throughput configuration) and the fusion tables (d = 64) use DIFFERENT state configurations and must not be compared numerically.

A.1 Theorem A(a) and A(b)

Fix a fibre v. The decoder observes (v, s) with s in (0,1)^B, so F(v, .) has at most 2^B distinct values. Let A(v) be the set of distinct answers over instances compatible with v and in the support. If |A(v)| > 2^B, two instances G, G' in the fibre share (v, s) but have k_H(G) != k_H(G'). F cannot equal both. Under the uniform measure on the fibre, at most 2^B of the |A(v)| answer classes can be correct, hence acc(F | v) <= 2^B / |A(v)|; this proves (a). The <= 1/2 corollary follows only when 2^B <= |A(v)|/2, i.e. B <= log2|A(v)| - 1. Exactness on every instance requires |A(v)| <= 2^B for all v, i.e. B >= log2 |A(v)|; this proves (b). Neither step uses A3. The marginal accuracy E_v[acc(F|v)] is NOT bounded by 1/|A| when fibres with |A(v)|=1 carry most of the mass, which is why Section 4 labels every statistic. QED.

Binary-chain witness for |A| = 2^m. Fix m <= n and H = m. At hop h the visible flat sequence contains one key k_h and TWO candidate successor symbols; a bit d_h not recorded in v selects which one is bound, and the selected symbol is the key of hop h+1. Both candidates are present as tokens, so every d in (0,1)^m yields the SAME v and the same q = k_1. Map d to the endpoint id sum_h d_h 2^(h-1); with V_K >= 2^m these are 2^m distinct endpoints. The witness enumerates all 2^m assignments with bit-identical visible inputs (max deviation 0.0e+00). If the pairing is also absent from v, all n! matchings belong to the fibre, giving B >= log2(n!). QED.

A.2 Theorem A(c) and the Toeplitz equivalence (P6)

In class M, layer h computes z_h = sum_t K_h(L-1-t) m_h[t] e(x_t) with m_h[t] = <q_h, e(x_(t-1))>, and q_(h+1) = T_h z_h / ||T_h z_h||. K_h depends only on the lag and on fixed parameters, T_h is fixed, and e is fixed. The composition is therefore a function of the multiset of (flat symbol id, position) pairs plus q, i.e. of v with the refinement s removed. Under arbitrary filler spacing the positions of the pair tokens are determined by filler counts independent of the role assignment, so F is constant on the fibre: no choice of H_L, state order or (rho, theta, w) changes this, and A(a) applies. To encode roles in the kernel one makes the recurrence coefficients functions of x_t, hence time-varying. For y_t = a_t y_(t-1) + b_t y_(t-2) + c_t u_t with c_t != 0, P6 gives h(s,s)=c_s, h(s+1,s)=a_(s+1) c_s, h(s+2,s)=(a_(s+2)a_(s+1)+b_(s+2)) c_s. Toeplitz forces c_s, a_s and b_s constant by induction on the lag, so (rho_t, theta_t) are constant. Non-Toeplitz h(t,s) has no single H(z) and the single-FFT path fails. This negates THAT path within M, not every conceivable O(L log L) algorithm. A(c) is therefore a conditional statement: IF duality is preserved, the bits must enter on the input side. QED.

A.3 Theorem A(d), A(e), A(f) and P2

A(d). By the Section 2 definition B = sum_t log2 |pi^(-1)(v_t)|, so an alphabet partition with |pi^(-1)| = 2 contributes exactly one bit per non-empty token and is COUNTED in B. An explicit tag and a symbol id carrying a role are the same quantity in a different carrier. Tagging only the KEY token of each of n pairs gives B = n, which attains B >= n from A(b). A positional prior is a support restriction: it reduces |A(v)| rather than increasing B, and under random filler spacing the restriction no longer holds, so the effective information falls to zero (measured 0.3438, 5 seeds). QED.

P2 (explicit construction). Fix keys and q; A = sum_p L_p v_p + c with d x d blocks L_p, read-out f(A,q) arbitrary, target j = j(q). Case 1: L_j singular. Pick nonzero delta in ker L_j and change only v_j; delta-A = 0, answer changes. Case 2: L_j invertible and some L_p != 0 for p != j. Pick delta_p and set delta_j = -L_j^(-1) L_p delta_p; then delta-A = L_p delta_p + L_j delta_j = 0, and delta_j is nonzero, so the target value changes. Case 3: L_j invertible and all other L_p = 0. Then A depends only on v_j; fix all values and redirect q to a different pair p != j, which changes the required answer while A is unchanged. In every case two legal assignments give the same (A,q) and different answers, so f cannot be exact. No step infers a target-coordinate perturbation from dim ker L > 0 alone. Scope: single mixer with a fixed linear historical summary; deep stacks with per-position nonlinear FFNs, input-dependent gates and outer-product states leave the premise. QED.

A(e). B < log2|A| makes exactness impossible for EVERY deterministic model by A(b), hence for every H_L; for B >= log2|A| Result 2(ii), under its stated premises, gives minimum depth H. The frontier is a right angle. A(f): under A4 the decoded value at depth H_L is k_(H_L); for H_L > H and distinct keys this differs from k_H. Retaining and selecting the H-th intermediate output gives the correct answer for any H_L >= H; the lower bound H_L >= H is unchanged. QED.

A.4 Result 2: induction and tightness

(i) P2 rules out an order-1 vector state with a linear summary; Construction C gives an order-2 tensor state with a contraction read-out, so the minimum is exactly 2. (ii) Induction on H. Base H=1: one contraction resolves one content match and yields the value; a zero-th layer cannot. Step: suppose H-1 layers resolve H-1 hops and produce q_H = C[k_H] by the unique H6 interface. Layer H performs one further match with q_H and returns k_H's successor. Fewer than H layers leave at least one hop unresolved: the last read-out sees a mixture of at least two successors and cannot be exact on all fibres. The analytic construction attains 1.0000 on the diagonal of the 5 x 5 grid and <= 0.10 off it, so the bound is tight and non-vacuous. Overshoot H_L > H is A(f), not a failure of the lower bound. QED.

A.5 Proposition P5: conditional state-order lower bound

Premise. The interface is H6 with rank at most D; the codebook rows are mutually orthogonal; at each of H hops there are c mutually orthogonal candidate successors; answers are determined by the joint choice of the start key and the H successors, so the candidate set has size c^(H+1). Claim. An order-o tensor state whose per-layer linear summary has image dimension at most o*d cannot be exact unless o >= H+1. Proof. The H+1 independent matching directions must be simultaneously recoverable from the summary for every one of the c^(H+1) joint choices. If o <= H, the summary is a linear map from a space of dimension (H+1)d into one of dimension o*D <= H*D; its kernel has dimension at least d > 0. By the P2 construction (with the blocks taken to be the coefficient matrices of the H+1 directions) there exist two distinct joint choices producing the same summary and the same q, hence the same read-out, yet requiring different answers. Exactness is impossible; therefore o >= H+1. This is a CONDITIONAL bound: when each hop offers only 2 candidates the same counting gives 2^(H+1) classes and the necessary order is weaker, degenerating to a log2(H+1) bit requirement. We state that degeneration explicitly rather than presenting c^(H+1) as unconditional. QED.

A.6 P1, P3, P4, P7, P8 and non-vacuity

P1. H(z) = sin(theta) / (1 - 2 rho cos(theta) z^(-1) + rho^2 z^(-2)) gives y_t = 2 rho cos(theta) y_(t-1) - rho^2 y_(t-2) + sin(theta) u_t, a real-field equivalence; different rounding orders in the FFT and IIR paths make fp32 bitwise identity false (W1). P3. For a finite output set F with |F| <= |S|, the state can encode f(x) directly, so the literal claim 'O(1) states must lose' is false; the correct statement is that a SPECTRAL compression state loses

content addressing. P4. At rho=1 and a single pole the Construction C state reduces to sum_tau k_tau v_tau^T, the linear-attention state; rho<1 retains a decay spectrum. P7. With impulse response bounded by rho^lag / |sin(theta)| and per-step injection bounded by delta_max, the output error is at most delta_max / ((1-rho)|sin(theta)|), under a uniform floating-point perturbation model. P8. As theta_m -> 0 the dictionary columns approach theta_m rho^lag (lag+1) and become collinear, so the grid is asymptotically rank deficient. Non-vacuity: n=4, m=2, V_K=32, H=2, flat alphabet, arbitrary filler spacing give |A|=4 at B=0 with acc <= 1/4, and exactness at B=2.

A.7 SE1: orthogonal-registration zero-crosstalk identity

PREMISE. C in R^(V x V) satisfies C C^T = I; S = sum_(a=1)^m w_a C[k_a]^T C[v_a] with distinct k_a; the query is q = C[k]. CLAIM. C[k] S = w_j C[V_K + v_j] when k = k_j, and 0 when k is not among the k_a. PROOF. C[k] C[k_a]^T = delta_(k,k_a) by row orthogonality, so C[k] S = sum_a w_a delta_(k,k_a) C[V_K+v_a]. Distinctness leaves at most one nonzero term. CONSEQUENCE. The read-out logits = (C[k] S) C[V_K:,:]^T are a one-hot vector scaled by w_j for a registered key, so accuracy is IDENTICALLY the fraction of correctly registered bindings. There is no cross-talk term to bound or estimate. NON-VACUITY. V = 64, V_K = 32, C the normalized Sylvester matrix, m >= 2: 200 randomized instances give a maximum on-target error of 4.4e-16 and maximum off-target error of 8.5e-17. BOUNDARY. Repeated keys, a rank-deficient codebook, or a non-codebook registration vector break the identity; this is why the break-alignment arm collapses to chance.

A.8 SE2: monotone improvement and cover time

PREMISE. n bindings arrive by i.i.d. sampling with replacement (coupon collector); the gate has false-accept at most alpha and miss at most beta; accepted experiences are correct; rho < 1 with a precision floor epsilon such that SE5's capacity K* >= n. CLAIM. (i) the number m_t of correct registrations is monotone non-decreasing; (ii) E[m_T] >= (1-beta) T until saturation; (iii) E[T_cover] = n H_n / (1-beta) with H_n = sum_(i=1)^n 1/i; (iv) Pr(T_cover > (1+eps) E T) <= exp(-eps^2 (1-beta) n H_n / 3); (v) the probability of any false commit in T rounds is at most min(1, T alpha). PROOF. (i) SE1 makes each correct registration a disjoint orthogonal term; register() replaces only the term of the same key and never removes another key's correct term, and K* >= n prevents decay below epsilon. (ii) The gate accepts a true binding with probability at least 1-beta. (iii) The classical coupon-collector expectation n H_n divided by the effective arrival probability 1-beta. (iv) Chernoff on a sum of n geometric variables, equivalently the union bound Pr(some type missing) <= n(1-1/n)^T. (v) Boole's inequality over T rounds. NON-VACUITY AND CHECK. For n = 32, alpha = 0.05, beta approximately 0, the prediction is 129.87 rounds; the measured five-seed mean first full-cover round is 130.00 (relative gap 0.10 percent), with per-seed values 170, 100, 90, 120, 170. With n = 5 the sample SD is 38.08 and the 95% t-CI is [82.72, 177.28] rounds: the prediction lies inside the interval, but the interval spans 73% of the mean, so this is weak evidence and the small gap is NOT a precision claim. The arrival model was set to i.i.d. BEFORE measuring. BOUNDARY. A deterministic round-robin schedule gives T_cover = n and SE2 does not apply; large beta stretches the cover time by 1/(1-beta); if K* < n then SE5 takes over and full cover is impossible.

A.9 SE3: the statistical gate is necessary

PREMISE. Each accepted experience is corrupted independently with probability p; a corrupted value is uniform on [0, V_K); there is no gate; the final value of a key is its last write. CLAIM. E[acc] = (1-p) + p/V_K = 1 - p(1 - 1/V_K) exactly, and E[acc] <= that value in general because some keys may never arrive. PROOF. The last write is correct either when it is not corrupted (probability 1-p) or when corruption happens to draw the true value (probability p/V_K). SE1 makes the keys independent, so accuracy is the mean of n such Bernoulli variables. CONCENTRATION. Over N independent key-level trials, Hoeffding gives Pr(|a_bar - mu| >= t) <= 2 exp(-2 N t^2). With N = 640000 the 4-sigma interval is [0.7071, 0.7116] and the enumeration mean is 0.7091, inside it. WHY NECESSARY. Orthogonality prevents cross-talk but cannot prevent a wrong value from occupying an orthogonal slot; hence without a gate SE2's full cover is unattainable under corruption. SCOPE. With p = 0, mu = 1 and the gate is irrelevant: the ungated clean arm ties the fused arm exactly. The gate is necessary ONLY when experiences can be wrong. W15. An earlier draft called mu an almost-sure upper bound and tested it pointwise against a sample mean; a sample mean legitimately exceeds an expectation. The claim is now an exact expectation with a concentration interval.

A.10 SE4: compatibility with the CT7 rate bound

CT7(b) bounds the information that T consolidation rounds can write into the state by T w b bits, independent of context length L and of bank capacity C. SE2 permits about (1-beta) T registrations and a corresponding accuracy gain. CLAIM. Both hold together. PROOF. Registration is injective (SE1's partial isometry), so one binding corresponds to one key-value association of exactly log2(V_K) bits. With T = 100 and w = 1, SE2 registers 100 bindings using 500 bits, while CT7 allows 500 bits: equality, not violation. MEANING. Self-evolution is real but bounded. Long context buys inference-time retrieval, not a faster learning rate. CT7(c)'s earlier statement that improvement was not proved is resolved for this finite process; CT7(b)'s upper bound is untouched.

A.11 SE5: optimal decay eviction and closed-form capacity

PREMISE. Slot weights w_a = rho^age(a) with rho < 1; utility is additive and orthogonal (SE1); precision floor epsilon; capacity C_max. CLAIM. (i) Greedy removal of the smallest weight is optimal; (ii) the number of slots whose weight stays above epsilon is K* = floor(log epsilon / log rho). PROOF. (i) Utility U = sum_a w_a u_a with u_a >= 0 and mutually independent; retaining the C_max largest weights maximizes U, and dropping the smallest is its online realization. (ii) rho^a > epsilon iff a < log epsilon / log rho. NON-VACUITY. At rho = 0.999 and epsilon = 1e-3, K* = 6904; at rho = 0.90, K* = 65. The fast-decay arm loses accuracy to 0.1313 while the standard arm stays at 1.0000 over 320 rounds, matching the predicted capacity collapse. BOUNDARY. rho = 1 is critically stable, non-BIBO and has amplitude growing with L; the architecture excludes it and no run instantiates it. A near-critical value such as 0.99999 barely decays within the run and therefore has no discriminating power; 0.90 was chosen so SE5 is observable.

Appendix B Withdrawn Statements

ID	Withdrawn statement and reason
W1	Unconditional fp32 bitwise FFT/IIR duality: different rounding orders; real-field equivalence only.
W8	Content addressing requires input-dependent kernel coefficients: refuted by Construction C with strictly static kernels.
W9	Absolute-position parity is a general role basis: a hidden format prior; collapses under arbitrary spacing.
W10/W1 1	'Exactly one bit per token is necessary and sufficient': the necessary quantity is total B >= log2|A|; one bit per token is a sufficient carrier.
W12	Single-hop attention parity (Fisher p=0.4998): the stored baseline did not reproduce in the same-run audit (Section 4.6). No equivalence or superiority claim is made.
W13	Trained corroboration of the layer bound under the earlier protocol: withdrawn after the fairness audit removed two input asymmetries. Re-established on the corrected architecture in Section 4.12; the earlier number is not reused.
W14	Straight-through inter-layer normalisation (norm taken outside the graph): finite-difference gradient deviation 2.5e-03. Replaced by a differentiable square root built from log and exp, deviation 2.9e-10; all conservation checks are rerun on final weights.
W15	SE3 as an almost-sure upper bound tested pointwise against a sample mean: a finite sample mean legitimately exceeds an expectation. Replaced by the exact expectation 1-p(1-1/V_K) with a Hoeffding concentration interval (Appendix A.9).

Appendix C Reproducibility

The submission is accompanied by an anonymised artifact containing the reference implementation of the layer, the seven same-run mechanism references used only for the Section 4.6 audit, the reachability enumeration

of Section 4.3, and all raw result files. Every number in Tables 2-4 is injected from those files at build time; no value is transcribed by hand. All probes are CPU-only NumPy and complete in minutes. The Section 4.12 run uses 1500 Adam steps, lr 0.02, batch 64, five seeds, and 512 training plus 512 held-out instances per seed; the architecture module and every convergence curve are in the artifact. The fusion run of Section 4.13 and Appendices D-E uses 320 i.i.d. arrival rounds, 32 bindings, five seeds, rho = 0.999 and epsilon = 1e-3; all seven arms, sixteen ablation cells and every individual seed are printed in Appendix D. Seeds are listed per table. The environment, dependency versions and a one-command reproduction script are in the artifact. The 2027 call for papers and submission deadline were not published at the time of writing; page count, the author-side LLM-use disclosure and reciprocal-reviewer obligations must be rechecked against the official template when it appears.

No count of statements is presented as a contribution. The withdrawn-statement register in Appendix B is a soundness practice, not an originality claim.

Appendix D All Raw Fusion Measurements

Every row below is a single seed of a single arm. Nothing is averaged away and no value is transcribed by hand: the generator reads the raw result files at build time. Protocol: 32 bindings, 320 rounds of i.i.d. arrival, rho = 0.999 unless stated, precision floor epsilon = 1e-3, capacity 64, one-sided paired-CI gate with alpha = 0.05, tau = 0, 8 probes and noise sigma = 0.10. Chance is 1/32 = 0.0313. The core state is 32768 bytes in every row (d = 64, fp64; notation table in Section 2). A machine-readable long-form CSV of every row below is supplied in the artifact as appendix_D_fusion_long.csv; no value is OCR-transcribed.

arm	seed	final acc	commits	first full cover
fused gated	0	1.0000	49	170
fused gated	1	1.0000	55	100
fused gated	2	1.0000	47	90
fused gated	3	1.0000	53	120
fused gated	4	1.0000	54	170
no consolidation	0	0.0000	0	none
no consolidation	1	0.0000	0	none
no consolidation	2	0.0000	0	none
no consolidation	3	0.0000	0	none
no consolidation	4	0.0000	0	none
ungated clean	0	1.0000	320	170
ungated clean	1	1.0000	320	100
ungated clean	2	1.0000	320	90
ungated clean	3	1.0000	320	120
ungated clean	4	1.0000	320	170
ungated corrupt	0	0.6250	320	none
ungated corrupt	1	0.7500	320	none
ungated corrupt	2	0.5625	320	none
ungated corrupt	3	0.5625	320	none
ungated corrupt	4	0.7188	320	none
random write	0	0.6250	154	none
random write	1	0.6875	153	none
random write	2	0.5938	154	none
random write	3	0.7500	155	none
random write	4	0.7812	155	none
break alignment	0	0.0000	49	none
break alignment	1	0.0000	55	none
break alignment	2	0.0938	47	none
break alignment	3	0.0000	53	none
break alignment	4	0.0312	54	none
rho fast	0	0.1562	49	none
rho fast	1	0.1562	55	none
rho fast	2	0.0938	47	none
rho fast	3	0.1250	53	none
rho fast	4	0.1250	54	none

Table 18. All 35 arm-seed runs. The first-full-cover column reports the first sampled round at which accuracy reaches 1.0 (sampling every 10 rounds); '-' means no such round occurred. The five fused values are the ones averaged in Section 4.13.

corruption	config	mean	sd	n
0.00	gate off	1.0000	0.0000	5
0.00	alpha 0.10	1.0000	0.0000	5
0.00	alpha 0.05	1.0000	0.0000	5
0.00	alpha 0.01	1.0000	0.0000	5
0.00	tau positive	1.0000	0.0000	5
0.00	sigma 0.30	1.0000	0.0000	5
0.00	rho 0.90	0.1313	0.0234	5
0.00	rho 0.999	1.0000	0.0000	5
0.30	gate off	0.6438	0.0781	5
0.30	alpha 0.10	1.0000	0.0000	5
0.30	alpha 0.05	1.0000	0.0000	5
0.30	alpha 0.01	1.0000	0.0000	5
0.30	tau positive	1.0000	0.0000	5
0.30	sigma 0.30	1.0000	0.0000	5
0.30	rho 0.90	0.1000	0.0364	5
0.30	rho 0.999	1.0000	0.0000	5

Table 19. All 16 corruption x configuration cells, each over five seeds. Under zero corruption the six gate configurations saturate and only the fast-decay configuration degrades; under 30 percent corruption the gate-off row degrades while every gated row saturates. This is the measured form of SE3's scope.

check	pass
theory 7of7	yes
theory SE1	yes
theory SE2	yes
theory SE3	yes
theory SE4	yes
theory SE5	yes
theory nonvacuous SE2	yes
theory SE3 concentration	yes
theory SE3 exact expectation	yes
evolution all invariants	yes
evolution monotone	yes
fused above 0.90	yes
all key paired positive	yes
core state bytes constant across rounds	yes
slot count bounded by capacity	yes
SE5 capacity predicts rho fast loss	yes
ablation clean gate configs saturate	yes
ablation noisy gate necessary	yes
ablation rho decay matches SE5	yes
SE2 prediction matches measurement	yes
SE3 concentration holds	yes
SE3 enumeration within 4sigma	yes
no gradient through retrieval	yes
no NxN score	yes
no kv cache	yes
v64 stack invariants	yes
v64 interface identities	yes

Table 20. All 27 consistency checks on the FINAL post-evolution state. They cover the five analytic results, the paired statistics, monotonicity, core-state constancy, capacity bounds, the SE2 prediction-measurement agreement, SE3's concentration interval, the v6.4 stack invariants and the interface identities. A false here would be a finding, not a crashed run.

Appendix E Premise, Boundary and Non-Vacuity Ledger

A theorem whose premises admit no instance is empty. The table below gives one concrete satisfying instance, the exact boundary at which the statement stops, and what would falsify it, for every new supporting result. These are not additional contributions; they are the audit trail that keeps the premises honest.

result	satisfying instance	boundary / falsifier
SE1	V=64, V_K=32, Sylvester C, m>=2 distinct keys	repeated keys or rank-deficient code; any nonzero off-target read
SE2	n=32 i.i.d. arrivals, beta~0, K*=6904 >= n	round-robin arrival; beta>0; K*<n; a non-monotone fused curve
SE3	p=0.30, uniform corrupted value, no gate	p=0 makes the gate irrelevant; a gated arm below the exact expectation
SE4	T=100, w=1, b=log2(32)=5, 500 bits used = 500 allowed	any run writing more than T w log2(V_K) bits into the state
SE5	rho=0.999, eps=1e-3, K*=6904; rho=0.90, K*=65	rho=1 is non-BIBO and excluded; a better-than-greedy eviction
CT7	answer() is no-grad; selection Jacobian vanishes a.e.	a differentiable exact-retrieval channel through the online path

Table 21. Each row names a measurement that would have to come out differently. The five supporting results SE1-SE5 do not change the four

core results of Section 1.1; they make the empirical claim in Result 4 auditable at the level of individual runs. The withdrawn-statement register (Appendix B) records W13, W14 and W15, all of which are premise corrections rather than numerical adjustments.

Appendix F All 1500-Step Convergence Curves

Section 4.12 reports only the diagonal and the first-crossing step. Here every sampled point of every arm is printed for all five seeds, followed by every individual cell of the layer-by-hop matrix. Protocol: 1500 Adam steps, lr 0.02, batch 64, 512 training and 512 held-out instances per seed, width d = 64, fixed Hadamard embedding, evaluation every 150 steps. The architecture module's inter-layer map is the fixed codebook value-to-key swap; only the query projection and read-out are learned, from random initialisation.

arm	step	mean acc	sd	n seeds
arch codebook M	0	0.0266	0.0064	5
arch codebook M	150	1.0000	0.0000	5
arch codebook M	300	1.0000	0.0000	5
arch codebook M	450	1.0000	0.0000	5
arch codebook M	600	1.0000	0.0000	5
arch codebook M	750	1.0000	0.0000	5
arch codebook M	900	1.0000	0.0000	5
arch codebook M	1050	1.0000	0.0000	5
arch codebook M	1200	1.0000	0.0000	5
arch codebook M	1350	1.0000	0.0000	5
arch codebook M	1500	1.0000	0.0000	5
arch analytic witness	0	1.0000	0.0000	5
legacy learnable interface random	0	0.0301	0.0052	5
legacy learnable interface random	150	0.2812	0.0368	5
legacy learnable interface random	300	0.2633	0.0381	5
legacy learnable interface random	450	0.2492	0.0319	5
legacy learnable interface random	600	0.2398	0.0278	5
legacy learnable interface random	750	0.2383	0.0388	5
legacy learnable interface random	900	0.2422	0.0276	5
legacy learnable interface random	1050	0.2445	0.0321	5
legacy learnable interface random	1200	0.2520	0.0329	5
legacy learnable interface random	1350	0.2535	0.0311	5
legacy learnable interface random	1500	0.2492	0.0325	5
legacy learnable interface fixedM	0	0.0387	0.0119	5
legacy learnable interface fixedM	150	1.0000	0.0000	5
legacy learnable interface fixedM	300	1.0000	0.0000	5
legacy learnable interface fixedM	450	1.0000	0.0000	5
legacy learnable interface fixedM	600	1.0000	0.0000	5
legacy learnable interface fixedM	750	1.0000	0.0000	5
legacy learnable interface fixedM	900	1.0000	0.0000	5
legacy learnable interface fixedM	1050	1.0000	0.0000	5
legacy learnable interface fixedM	1200	1.0000	0.0000	5
legacy learnable interface fixedM	1350	1.0000	0.0000	5
legacy learnable interface fixedM	1500	1.0000	0.0000	5
legacy trainable near-swap	0	0.0238	0.0106	5
legacy trainable near-swap	150	1.0000	0.0000	5
legacy trainable near-swap	300	1.0000	0.0000	5
legacy trainable near-swap	450	1.0000	0.0000	5
legacy trainable near-swap	600	1.0000	0.0000	5
legacy trainable near-swap	750	1.0000	0.0000	5
legacy trainable near-swap	900	1.0000	0.0000	5
legacy trainable near-swap	1050	1.0000	0.0000	5
legacy trainable near-swap	1200	1.0000	0.0000	5
legacy trainable near-swap	1350	1.0000	0.0000	5
legacy trainable near-swap	1500	1.0000	0.0000	5

Table 22. All 45 arm-step cells. The random learnable interface stays in 0.21-0.29 for the entire budget (this task's chance is 1/32 = 0.0313; 0.25 belongs to a DIFFERENT historical task); the codebook-fixed interface and the near-swap basin cross 0.99 by step 150 on every seed. The analytic witness is exact at step 0 by construction.

hops	layers	seed	acc	params	steps
1	1	0	1.0000	6144	1500
1	1	1	1.0000	6144	1500
1	1	2	1.0000	6144	1500
1	1	3	1.0000	6144	1500
1	1	4	1.0000	6144	1500
1	2	0	0.0215	6144	1500
1	2	1	0.0430	6144	1500
1	2	2	0.0332	6144	1500
1	2	3	0.0312	6144	1500
1	2	4	0.0273	6144	1500
1	3	0	0.0352	6144	1500
1	3	1	0.0410	6144	1500
1	3	2	0.0312	6144	1500
1	3	3	0.0312	6144	1500
1	3	4	0.0566	6144	1500
1	4	0	0.0469	6144	1500
1	4	1	0.0410	6144	1500
1	4	2	0.0352	6144	1500
1	4	3	0.0312	6144	1500
1	4	4	0.0273	6144	1500
2	1	0	0.1582	6144	1500
2	1	1	0.1543	6144	1500
2	1	2	0.2129	6144	1500
2	1	3	0.1875	6144	1500
2	1	4	0.1914	6144	1500
2	2	0	1.0000	6144	1500
2	2	1	1.0000	6144	1500
2	2	2	1.0000	6144	1500
2	2	3	1.0000	6144	1500
2	2	4	1.0000	6144	1500
2	3	0	0.0352	6144	1500
2	3	1	0.0254	6144	1500
2	3	2	0.0273	6144	1500
2	3	3	0.0312	6144	1500
2	3	4	0.0312	6144	1500
2	4	0	0.0352	6144	1500
2	4	1	0.0410	6144	1500
2	4	2	0.0371	6144	1500
2	4	3	0.0273	6144	1500
2	4	4	0.0332	6144	1500
3	1	0	0.1562	6144	1500
3	1	1	0.1660	6144	1500
3	1	2	0.2051	6144	1500
3	1	3	0.1230	6144	1500
3	1	4	0.1348	6144	1500
3	2	0	0.2363	6144	1500
3	2	1	0.2480	6144	1500
3	2	2	0.2500	6144	1500
3	2	3	0.2773	6144	1500
3	2	4	0.2246	6144	1500
3	3	0	1.0000	6144	1500
3	3	1	1.0000	6144	1500
3	3	2	1.0000	6144	1500
3	3	3	1.0000	6144	1500
3	3	4	1.0000	6144	1500
3	4	0	0.0352	6144	1500
3	4	1	0.0449	6144	1500
3	4	2	0.0332	6144	1500
3	4	3	0.0352	6144	1500
3	4	4	0.0332	6144	1500
4	1	0	0.1348	6144	1500
4	1	1	0.1875	6144	1500
4	1	2	0.1719	6144	1500
4	1	3	0.1504	6144	1500
4	1	4	0.1309	6144	1500
4	2	0	0.2402	6144	1500
4	2	1	0.2051	6144	1500
4	2	2	0.2637	6144	1500
4	2	3	0.2773	6144	1500
4	2	4	0.2207	6144	1500
4	3	0	0.3555	6144	1500
4	3	1	0.3516	6144	1500
4	3	2	0.3496	6144	1500
4	3	3	0.3633	6144	1500
4	3	4	0.3535	6144	1500
4	4	0	1.0000	6144	1500
4	4	1	1.0000	6144	1500
4	4	2	1.0000	6144	1500
4	4	3	1.0000	6144	1500
4	4	4	1.0000	6144	1500

Table 23. All 80 layer-by-hop runs. Parameter count is 6144 in every row because the inter-layer map is shared and is not a parameter; depth adds computation, not weights. The diagonal is the only set of cells above 0.99.

Appendix G All Fusion Evolution Curves

Section 4.13 reports final accuracies. Here the full trajectory of each arm is printed: accuracy is sampled every 10 rounds and averaged over the five seeds at each sampled round. Protocol: 32 bindings, 320 i.i.d. arrival rounds, rho = 0.999 except the fast-decay arm, epsilon = 1e-3, capacity 64, one-sided paired-CI gate. The core state is 32768 bytes at every sampled point of every run.

arm	round	mean acc	sd	mean slots
fused gated	1	0.0312	0.0000	1.0
fused gated	10	0.2687	0.0250	8.6
fused gated	20	0.4688	0.0198	15.0
fused gated	30	0.6562	0.0685	21.0
fused gated	40	0.7438	0.0459	23.8
fused gated	50	0.7812	0.0442	25.0
fused gated	60	0.8562	0.0153	27.4
fused gated	70	0.9000	0.0415	28.8
fused gated	80	0.9125	0.0415	29.2
fused gated	90	0.9500	0.0375	30.4
fused gated	100	0.9625	0.0364	30.8
fused gated	110	0.9750	0.0234	31.2
fused gated	120	0.9812	0.0250	31.4
fused gated	130	0.9812	0.0250	31.4
fused gated	140	0.9875	0.0153	31.6
fused gated	150	0.9875	0.0153	31.6
fused gated	160	0.9875	0.0153	31.6
fused gated	170	1.0000	0.0000	32.0
fused gated	180	1.0000	0.0000	32.0
fused gated	190	1.0000	0.0000	32.0
fused gated	200	1.0000	0.0000	32.0
fused gated	210	1.0000	0.0000	32.0
fused gated	220	1.0000	0.0000	32.0
fused gated	230	1.0000	0.0000	32.0
fused gated	240	1.0000	0.0000	32.0
fused gated	250	1.0000	0.0000	32.0
fused gated	260	1.0000	0.0000	32.0
fused gated	270	1.0000	0.0000	32.0
fused gated	280	1.0000	0.0000	32.0
fused gated	290	1.0000	0.0000	32.0
fused gated	300	1.0000	0.0000	32.0
fused gated	310	1.0000	0.0000	32.0
fused gated	320	1.0000	0.0000	32.0
no consolidation	1	0.0000	0.0000	0.0
no consolidation	10	0.0000	0.0000	0.0
no consolidation	20	0.0000	0.0000	0.0
no consolidation	30	0.0000	0.0000	0.0
no consolidation	40	0.0000	0.0000	0.0
no consolidation	50	0.0000	0.0000	0.0
no consolidation	60	0.0000	0.0000	0.0
no consolidation	70	0.0000	0.0000	0.0
no consolidation	80	0.0000	0.0000	0.0
no consolidation	90	0.0000	0.0000	0.0
no consolidation	100	0.0000	0.0000	0.0
no consolidation	110	0.0000	0.0000	0.0
no consolidation	120	0.0000	0.0000	0.0
no consolidation	130	0.0000	0.0000	0.0
no consolidation	140	0.0000	0.0000	0.0
no consolidation	150	0.0000	0.0000	0.0
no consolidation	160	0.0000	0.0000	0.0
no consolidation	170	0.0000	0.0000	0.0
no consolidation	180	0.0000	0.0000	0.0
no consolidation	190	0.0000	0.0000	0.0
no consolidation	200	0.0000	0.0000	0.0
no consolidation	210	0.0000	0.0000	0.0
no consolidation	220	0.0000	0.0000	0.0
no consolidation	230	0.0000	0.0000	0.0
no consolidation	240	0.0000	0.0000	0.0
no consolidation	250	0.0000	0.0000	0.0
no consolidation	260	0.0000	0.0000	0.0
no consolidation	270	0.0000	0.0000	0.0
no consolidation	280	0.0000	0.0000	0.0
no consolidation	290	0.0000	0.0000	0.0
no consolidation	300	0.0000	0.0000	0.0
no consolidation	310	0.0000	0.0000	0.0
no consolidation	320	0.0000	0.0000	0.0
ungated clean	1	0.0312	0.0000	1.0
ungated clean	10	0.2687	0.0250	8.6
ungated clean	20	0.4688	0.0198	15.0
ungated clean	30	0.6562	0.0685	21.0
ungated clean	40	0.7438	0.0459	23.8
ungated clean	50	0.7812	0.0442	25.0
ungated clean	60	0.8562	0.0153	27.4
ungated clean	70	0.9000	0.0415	28.8
ungated clean	80	0.9125	0.0415	29.2
ungated clean	90	0.9500	0.0375	30.4
ungated clean	100	0.9625	0.0364	30.8
ungated clean	110	0.9750	0.0234	31.2
ungated clean	120	0.9812	0.0250	31.4
ungated clean	130	0.9812	0.0250	31.4
ungated clean	140	0.9875	0.0153	31.6
ungated clean	150	0.9875	0.0153	31.6
ungated clean	160	0.9875	0.0153	31.6
ungated clean	170	1.0000	0.0000	32.0
ungated clean	180	1.0000	0.0000	32.0
ungated clean	190	1.0000	0.0000	32.0
ungated clean	200	1.0000	0.0000	32.0
ungated clean	210	1.0000	0.0000	32.0
ungated clean	220	1.0000	0.0000	32.0
ungated clean	230	1.0000	0.0000	32.0
ungated clean	240	1.0000	0.0000	32.0
ungated clean	250	1.0000	0.0000	32.0
ungated clean	260	1.0000	0.0000	32.0
ungated clean	270	1.0000	0.0000	32.0
ungated clean	280	1.0000	0.0000	32.0
ungated clean	290	1.0000	0.0000	32.0
ungated clean	300	1.0000	0.0000	32.0
ungated clean	310	1.0000	0.0000	32.0
ungated clean	320	1.0000	0.0000	32.0
ungated corrupt	1	0.0125	0.0153	1.0
ungated corrupt	10	0.1750	0.0153	8.6
ungated corrupt	20	0.3125	0.0713	15.2
ungated corrupt	30	0.4250	0.0580	20.4
ungated corrupt	40	0.4875	0.0424	23.4
ungated corrupt	50	0.5813	0.0580	25.6
ungated corrupt	60	0.6188	0.0893	26.8
ungated corrupt	70	0.6188	0.0459	28.4
ungated corrupt	80	0.6375	0.0424	29.8
ungated corrupt	90	0.6750	0.0319	30.2
ungated corrupt	100	0.6813	0.0306	31.0
ungated corrupt	110	0.6750	0.0424	31.2
ungated corrupt	120	0.6937	0.0234	31.4
ungated corrupt	130	0.6750	0.0468	31.4
ungated corrupt	140	0.6937	0.0606	31.6
ungated corrupt	150	0.7125	0.0500	32.0
ungated corrupt	160	0.7500	0.0559	32.0
ungated corrupt	170	0.7125	0.0800	32.0
ungated corrupt	180	0.7125	0.0824	32.0
ungated corrupt	190	0.7188	0.1296	32.0
ungated corrupt	200	0.7188	0.1479	32.0
ungated corrupt	210	0.7562	0.1389	32.0
ungated corrupt	220	0.7438	0.0956	32.0
ungated corrupt	230	0.7312	0.1019	32.0
ungated corrupt	240	0.7500	0.0625	32.0
ungated corrupt	250	0.6687	0.0673	32.0
ungated corrupt	260	0.6687	0.0755	32.0
ungated corrupt	270	0.6937	0.0415	32.0
ungated corrupt	280	0.7188	0.0713	32.0
ungated corrupt	290	0.7063	0.0153	32.0
ungated corrupt	300	0.6562	0.0198	32.0
ungated corrupt	310	0.6312	0.0234	32.0
ungated corrupt	320	0.6438	0.0781	32.0
random write	1	0.0000	0.0000	0.2
random write	10	0.1000	0.0125	4.8
random write	20	0.1250	0.0198	6.0
random write	30	0.1625	0.0459	7.8
random write	40	0.2500	0.0198	11.2
random write	50	0.3375	0.0125	15.0
random write	60	0.4062	0.0685	17.4
random write	70	0.4562	0.0805	20.6
random write	80	0.5062	0.0538	22.2
random write	90	0.5188	0.0468	22.6
random write	100	0.5250	0.0667	24.2
random write	110	0.5437	0.0545	25.0
random write	120	0.5750	0.0319	25.4
random write	130	0.5625	0.0625	26.0
random write	140	0.5750	0.0643	26.8
random write	150	0.6000	0.0637	27.0
random write	160	0.6375	0.1038	28.0
random write	170	0.6625	0.0871	28.8
random write	180	0.6750	0.0755	29.2
random write	190	0.7125	0.0935	29.4
random write	200	0.7000	0.1179	29.8
random write	210	0.7375	0.1000	30.0
random write	220	0.7250	0.0956	30.4
random write	230	0.7375	0.0897	30.8
random write	240	0.7500	0.0765	31.0
random write	250	0.6875	0.1046	31.2
random write	260	0.6875	0.1064	31.2
random write	270	0.7188	0.0948	31.4
random write	280	0.7063	0.0919	31.4
random write	290	0.7000	0.0805	31.4
random write	300	0.6813	0.0696	31.4

random write	310	0.6687	0.0508	31.4
random write	320	0.6875	0.0713	31.4
break alignment	1	0.0437	0.0250	1.0
break alignment	10	0.0312	0.0342	8.6
break alignment	20	0.0250	0.0125	15.0
break alignment	30	0.0312	0.0342	21.0
break alignment	40	0.0187	0.0250	23.8
break alignment	50	0.0187	0.0250	25.0
break alignment	60	0.0312	0.0198	27.4
break alignment	70	0.0312	0.0342	28.8
break alignment	80	0.0312	0.0342	29.2
break alignment	90	0.0312	0.0342	30.4
break alignment	100	0.0312	0.0395	30.8
break alignment	110	0.0125	0.0250	31.2
break alignment	120	0.0125	0.0250	31.4
break alignment	130	0.0063	0.0125	31.4
break alignment	140	0.0187	0.0153	31.6
break alignment	150	0.0125	0.0153	31.6
break alignment	160	0.0125	0.0153	31.6
break alignment	170	0.0250	0.0234	32.0
break alignment	180	0.0250	0.0234	32.0
break alignment	190	0.0187	0.0153	32.0
break alignment	200	0.0125	0.0153	32.0
break alignment	210	0.0125	0.0153	32.0
break alignment	220	0.0187	0.0250	32.0
break alignment	230	0.0312	0.0280	32.0
break alignment	240	0.0312	0.0280	32.0
break alignment	250	0.0250	0.0234	32.0
break alignment	260	0.0187	0.0250	32.0
break alignment	270	0.0187	0.0153	32.0
break alignment	280	0.0187	0.0153	32.0
break alignment	290	0.0125	0.0153	32.0
break alignment	300	0.0250	0.0364	32.0
break alignment	310	0.0250	0.0364	32.0
break alignment	320	0.0250	0.0364	32.0
rho fast	1	0.0312	0.0000	1.0
rho fast	10	0.2687	0.0250	8.6
rho fast	20	0.4688	0.0198	15.0
rho fast	30	0.6562	0.0685	21.0
rho fast	40	0.7438	0.0459	23.8
rho fast	50	0.7812	0.0442	25.0
rho fast	60	0.8562	0.0153	27.4
rho fast	70	0.7625	0.0755	28.8
rho fast	80	0.5813	0.0580	29.2
rho fast	90	0.4188	0.0897	30.4
rho fast	100	0.3250	0.1000	30.8
rho fast	110	0.3063	0.0893	31.2
rho fast	120	0.2750	0.0824	31.4
rho fast	130	0.2375	0.0755	31.4
rho fast	140	0.2313	0.0643	31.6
rho fast	150	0.2000	0.0755	31.6
rho fast	160	0.1750	0.0545	31.6
rho fast	170	0.1562	0.0656	32.0
rho fast	180	0.1562	0.0442	32.0
rho fast	190	0.1875	0.0523	32.0
rho fast	200	0.1500	0.0750	32.0
rho fast	210	0.1500	0.0871	32.0
rho fast	220	0.1500	0.0696	32.0
rho fast	230	0.1250	0.0656	32.0
rho fast	240	0.1375	0.0643	32.0
rho fast	250	0.1187	0.0306	32.0
rho fast	260	0.1125	0.0250	32.0
rho fast	270	0.1375	0.0153	32.0
rho fast	280	0.1250	0.0280	32.0
rho fast	290	0.1313	0.0364	32.0
rho fast	300	0.1437	0.0250	32.0
rho fast	310	0.1250	0.0342	32.0
rho fast	320	0.1313	0.0234	32.0

Table 24. All 231 arm-round cells. The fused arm is monotone on all five seeds; the ungated corrupted arm rises then fluctuates as later writes overwrite earlier correct registrations; the fast-decay arm loses slots as SE5's finite-precision capacity is exhausted.

comparison	paired diff	CI95 low	CI95 high	all pos
no consolidation	+1.0000	+1.0000	+1.0000	yes
ungated clean	+0.0000	+0.0000	+0.0000	no
ungated corrupt	+0.3563	+0.2875	+0.4250	yes
random write	+0.3125	+0.2500	+0.3750	yes
break alignment	+0.9750	+0.9375	+1.0000	yes
rho fast	+0.8688	+0.8500	+0.8875	yes

Table 25. All 6 paired bootstrap comparisons, 10000 resamples, five seeds. The ungated clean arm ties exactly: with no corruption the gate is irrelevant, which is the measured form of SE3's scope, not a failure of the comparison.

Appendix H Exhaustive Information-Bound Enumeration

Theorem A is checked against the complete enumeration of the binary-chain fibre space, not against a sample. Every (m, B) pair below is enumerated over all 2^m role assignments with bit-identical visible inputs. The depth scan then verifies that stacking 1-8 static-LTI layers produces bit-identical outputs across compatible role assignments. These are finite exhaustive objects; they certify non-vacuity, they are not the proof.

m	B	answer classes	fibres	acc	predicted
1	0	2	1	0.5000	0.5000
1	1	2	2	1.0000	1.0000
1	2	2	2	1.0000	1.0000
2	0	4	1	0.2500	0.2500
2	1	4	2	0.5000	0.5000
2	2	4	4	1.0000	1.0000
2	3	4	4	1.0000	1.0000
3	0	8	1	0.1250	0.1250
3	1	8	2	0.2500	0.2500
3	2	8	4	0.5000	0.5000
3	3	8	8	1.0000	1.0000
3	4	8	8	1.0000	1.0000
4	0	16	1	0.0625	0.0625
4	1	16	2	0.1250	0.1250
4	2	16	4	0.2500	0.2500
4	3	16	8	0.5000	0.5000
4	4	16	16	1.0000	1.0000
4	5	16	16	1.0000	1.0000

Table 26. All 18 (m, B) enumeration cells. Accuracy equals the predicted upper bound 2^B / |A| in every row and reaches 1 exactly at B = m.

m	B	layers	|A|	acc	output dev
2	0	1	4	0.2500	0.0e+00
2	0	2	4	0.2500	0.0e+00
2	0	3	4	0.2500	0.0e+00
2	0	4	4	0.2500	0.0e+00
2	0	5	4	0.2500	0.0e+00
2	0	6	4	0.2500	0.0e+00
2	0	7	4	0.2500	0.0e+00
2	0	8	4	0.2500	0.0e+00
2	1	1	4	0.5000	0.0e+00
2	1	2	4	0.5000	0.0e+00
2	1	3	4	0.5000	0.0e+00
2	1	4	4	0.5000	0.0e+00
2	1	5	4	0.5000	0.0e+00
2	1	6	4	0.5000	0.0e+00
2	1	7	4	0.5000	0.0e+00
2	1	8	4	0.5000	0.0e+00
2	2	1	4	1.0000	0.0e+00
2	2	2	4	1.0000	0.0e+00
2	2	3	4	1.0000	0.0e+00
2	2	4	4	1.0000	0.0e+00
2	2	5	4	1.0000	0.0e+00
2	2	6	4	1.0000	0.0e+00
2	2	7	4	1.0000	0.0e+00
2	2	8	4	1.0000	0.0e+00
3	0	1	8	0.1250	0.0e+00
3	0	2	8	0.1250	0.0e+00
3	0	3	8	0.1250	0.0e+00
3	0	4	8	0.1250	0.0e+00
3	0	5	8	0.1250	0.0e+00
3	0	6	8	0.1250	0.0e+00
3	0	7	8	0.1250	0.0e+00
3	0	8	8	0.1250	0.0e+00
3	1	1	8	0.2500	0.0e+00
3	1	2	8	0.2500	0.0e+00
3	1	3	8	0.2500	0.0e+00
3	1	4	8	0.2500	0.0e+00
3	1	5	8	0.2500	0.0e+00
3	1	6	8	0.2500	0.0e+00
3	1	7	8	0.2500	0.0e+00
3	1	8	8	0.2500	0.0e+00
3	3	1	8	1.0000	0.0e+00
3	3	2	8	1.0000	0.0e+00
3	3	3	8	1.0000	0.0e+00
3	3	4	8	1.0000	0.0e+00
3	3	5	8	1.0000	0.0e+00
3	3	6	8	1.0000	0.0e+00
3	3	7	8	1.0000	0.0e+00
3	3	8	8	1.0000	0.0e+00
4	0	1	16	0.0625	0.0e+00
4	0	2	16	0.0625	0.0e+00
4	0	3	16	0.0625	0.0e+00
4	0	4	16	0.0625	0.0e+00

4	0	5	16	0.0625	0.0e+00
4	0	6	16	0.0625	0.0e+00
4	0	7	16	0.0625	0.0e+00
4	0	8	16	0.0625	0.0e+00
4	1	1	16	0.1250	0.0e+00
4	1	2	16	0.1250	0.0e+00
4	1	3	16	0.1250	0.0e+00
4	1	4	16	0.1250	0.0e+00
4	1	5	16	0.1250	0.0e+00
4	1	6	16	0.1250	0.0e+00
4	1	7	16	0.1250	0.0e+00
4	1	8	16	0.1250	0.0e+00
4	4	1	16	1.0000	0.0e+00
4	4	2	16	1.0000	0.0e+00
4	4	3	16	1.0000	0.0e+00
4	4	4	16	1.0000	0.0e+00
4	4	5	16	1.0000	0.0e+00
4	4	6	16	1.0000	0.0e+00
4	4	7	16	1.0000	0.0e+00
4	4	8	16	1.0000	0.0e+00
1	1	1	2	1.0000	0.0e+00
1	1	2	2	0.0000	0.0e+00
1	1	3	2	0.0000	0.0e+00
1	1	4	2	0.0000	0.0e+00
1	1	5	2	0.0586	0.0e+00
2	2	1	4	0.0000	0.0e+00
2	2	2	4	1.0000	0.0e+00
2	2	3	4	0.0000	0.0e+00
2	2	4	4	0.0000	0.0e+00
2	2	5	4	0.0625	0.0e+00
3	3	1	8	0.0000	0.0e+00
3	3	2	8	0.0000	0.0e+00
3	3	3	8	1.0000	0.0e+00
3	3	4	8	0.0000	0.0e+00
3	3	5	8	0.0801	0.0e+00
4	4	1	16	0.0000	0.0e+00
4	4	2	16	0.0000	0.0e+00
4	4	3	16	0.0000	0.0e+00
4	4	4	16	1.0000	0.0e+00
4	4	5	16	0.0898	0.0e+00
2	2	3	4	1.0000	0.0e+00
2	2	4	4	1.0000	0.0e+00
3	3	4	8	1.0000	0.0e+00
3	3	5	8	1.0000	0.0e+00
4	4	5	16	1.0000	0.0e+00
4	4	6	16	1.0000	0.0e+00

Table 27. All 98 depth-scan cells. Accuracy and output deviation are invariant to layer count, which is the empirical form of Theorem A(e): depth cannot manufacture the missing bits.

carrier	bits / token	acc	Toeplitz	run-time tag
explicit tag	1	1.0000	yes	yes
codebook partition	1	1.0000	yes	no

Table 28. All 3 bit carriers. The position-format-prior accuracy 0.2969 is a SINGLE-SEED measurement on the alternating layout under that carrier's support assumption; it is a DIFFERENT configuration from the frozen five-seed parity-gapped value in Table 10 and is never quoted as a conclusion elsewhere. The position format prior is a support restriction, not a per-token channel, and is analysed through A(v) rather than through B; its effective bit count under arbitrary filler is zero.

hops	layers	acc	per-layer accuracies
1	1	1.0000	1.00
1	2	0.0000	1.00 0.00
1	3	0.0000	1.00 0.00 0.00
1	4	0.0000	1.00 0.00 0.00 0.00
1	5	0.0645	1.00 0.00 0.00 0.00 0.06
2	1	0.0000	0.00
2	2	1.0000	0.00 1.00
2	3	0.0000	0.00 1.00 0.00
2	4	0.0000	0.00 1.00 0.00 0.00
2	5	0.0684	0.00 1.00 0.00 0.00 0.07
3	1	0.0000	0.00
3	2	0.0000	0.00 0.00
3	3	1.0000	0.00 0.00 1.00
3	4	0.0000	0.00 0.00 1.00 0.00
3	5	0.0879	0.00 0.00 1.00 0.00 0.09
4	1	0.0000	0.00
4	2	0.0000	0.00 0.00
4	3	0.0000	0.00 0.00 0.00
4	4	1.0000	0.00 0.00 0.00 1.00
4	5	0.0918	0.00 0.00 0.00 1.00 0.09

Table 29. All 20 zero-training layer-by-hop cells with the full per-layer accuracy vector. The last-layer read-out is why H_L > H overshoots fail: the correct intermediate value is computed at layer H but is not the layer that is read out.

Appendix I Complete Numeric Provenance and Witness Detail

Every numeric value quoted anywhere in the paper is listed below with its statistic, seed count, sample count and the object it was measured on. This table exists so that no number has to be taken on trust and no earlier drifted value can re-enter unnoticed. All entries are finite-sample quantities unless the statistic column says otherwise; none is a theorem.

quantity	value	statistic	seeds / n	object
chance (this task)	0.0312	definition	-	1/V_K
H=2, 2 layers, 1500 steps	1.0000	mean / sd 0.0000	5 / 512	architecture stack
H=2, 1 layer, 1500 steps	0.1809	mean / sd 0.0219	5 / 512	architecture stack
H=2 paired difference	+0.8191	paired bootstrap	5	two-layer minus one-layer
paired 95%% CI	[+0.8008, +0.8375]	10000 resamples	5	same pair
first step >= 0.99	150	max over seeds	5	codebook-fixed interface
legacy random interface, 1500	0.2492	mean	5	learnable inter-layer matrix
SE2 predicted E[T_cover]	129.87	analytic n H_n / (1-beta)	-	n=32, beta~0
SE2 measured mean cover	130.00	mean over seeds	5	fused gated arm
SE2 relative gap	0.0987%	derived (mean coincidence, NOT precision)	-	prediction vs measurement
SE2 cover 95% t-CI	[82.72, 177.28]	t(4), sample SD 38.08	5	per-seed first full cover
SE3 exact E[acc]	0.709375	analytic	-	p=0.30, V_K=32
SE3 enumeration mean	0.709102	mean over 640000 key trials	20000 runs	exhaustive per-key
SE3 Hoeffding 4-sigma	[0.7071, 0.7116]	concentration	-	same
SE4 bits consumed	500	analytic	-	T=100, w=1, b=5
SE4 CT7 bound	500	analytic upper bound	-	same
SE5 K* at rho=0.999	6904	floor(log eps / log rho)	-	eps=1e-3
core state bytes	32768	exact	-	d x d fp64, all runs
fused final accuracy	1.0000	mean / sd 0.0000	5 / 512	320 rounds
ungated corrupted	0.6438	mean / sd 0.0781	5 / 512	p=0.30, no gate
break alignment	0.0250	mean / sd 0.0364	5 / 512	non-codebook registration
fast decay	0.1313	mean / sd 0.0234	5 / 512	rho=0.90
SE1 max on-target error	4.4e-16	max over 200 instances	200	orthogonality identity
SE1 max off-target error	8.5e-17	max over 200 instances	200	same
FFT/IIR duality (final state)	2.89e-16	max over probes	-	real-field equivalence
Toeplitz deviation (final)	0	exact	-	static recurrence

Table 30. All 26 provenance rows. Values from different probes are never averaged together; the FFT/IIR row is a maximum over separately listed probes, not a mixture.

witness	measured value	what it certifies
evolution all invariants	pass	consistency on final weights
evolution monotone	pass	consistency on final weights
fused above 0.90	pass	consistency on final weights
all key paired positive	pass	consistency on final weights
core state bytes constant across rounds	pass	consistency on final weights
slot count bounded by capacity	pass	consistency on final weights
SE5 capacity predicts rho fast loss	pass	consistency on final weights
ablation clean gate configs saturate	pass	consistency on final weights
ablation noisy gate necessary	pass	consistency on final weights
ablation rho decay matches SE5	pass	consistency on final weights
SE2 prediction matches measurement	pass	consistency on final weights
SE3 concentration holds	pass	consistency on final weights
SE3 enumeration within 4sigma	pass	consistency on final weights
no gradient through retrieval	pass	consistency on final weights
no NxN score	pass	consistency on final weights
no kv cache	pass	consistency on final weights
SE2 prediction gap	0.0987%	coupon-collector bound is non-vacuous
SE3 concentration	inside 4-sigma	exact expectation, not an upper bound
SE4 equality	500 = 500 bits	no contradiction with CT7
SE5 capacity	6904 vs 65	decay predicts the loss

Table 31. Witness detail for the non-analytic checks. A false here would be a finding to fix, not a crashed run.

hops	encoding	n	mean |A|	max log2|A|	frac |A|>1
1	flat	1280	1.00	0.00	0.00
1	codebook	1280	1.00	0.00	0.00
2	flat	1280	1.52	1.00	0.52
2	codebook	1280	1.00	0.00	0.00
3	flat	1280	2.36	2.00	0.75
3	codebook	1280	1.00	0.00	0.00
4	flat	1280	3.25	2.32	0.97
4	codebook	1280	1.00	0.00	0.00

Table 32. Compatible-answer-set enumeration for the original content-chain task. The bound B >= log2 |A(v)| is sufficient but not tight on this task; tightness belongs to the binary-chain witness of Appendix H. This distinction is stated in Section 4.3 and is not hidden here.

Appendix J Per-Seed Raw Curves at Key Steps

Appendix F averages over seeds at each step; this appendix prints the individual trajectories so that no seed-level failure can be hidden by a mean. Each row is one seed of one arm at one evaluation step. The five evaluation steps are 0, 150, 400, 800 and 1500, taken from the same 1500-step runs used in Section 4.12. Loss is the held-out cross-entropy on a fixed 64-instance probe batch; accuracy is on the full 512-instance held-out set.

arm	seed	step	acc	loss
arch codebook M	0	0	0.0273	3.4683
arch codebook M	0	150	1.0000	0.0655
arch codebook M	0	1500	1.0000	0.0017
arch codebook M	1	0	0.0273	3.4534
arch codebook M	1	150	1.0000	0.0608
arch codebook M	1	1500	1.0000	0.0015
arch codebook M	2	0	0.0234	3.4731
arch codebook M	2	150	1.0000	0.0597
arch codebook M	2	1500	1.0000	0.0014
arch codebook M	3	0	0.0371	3.4595
arch codebook M	3	150	1.0000	0.0751
arch codebook M	3	1500	1.0000	0.0024
arch codebook M	4	0	0.0176	3.4635
arch codebook M	4	150	1.0000	0.0756
arch codebook M	4	1500	1.0000	0.0021
arch analytic witness	0	0	1.0000	0.0000
arch analytic witness	1	0	1.0000	0.0000
arch analytic witness	2	0	1.0000	0.0000
arch analytic witness	3	0	1.0000	0.0000
arch analytic witness	4	0	1.0000	0.0000
legacy learnable interface rando	0	0	0.0332	3.4666
legacy learnable interface rando	0	150	0.2852	3.8409
legacy learnable interface rando	0	1500	0.2617	4.9533
legacy learnable interface rando	1	0	0.0293	3.4664
legacy learnable interface rando	1	150	0.2637	4.0069
legacy learnable interface rando	1	1500	0.2129	5.6684
legacy learnable interface rando	2	0	0.0371	3.4607
legacy learnable interface rando	2	150	0.2207	3.6956
legacy learnable interface rando	2	1500	0.2090	4.7915
legacy learnable interface rando	3	0	0.0215	3.4631
legacy learnable interface rando	3	150	0.3223	3.6256
legacy learnable interface rando	3	1500	0.2734	6.1103
legacy learnable interface rando	4	0	0.0293	3.4697
legacy learnable interface rando	4	150	0.3145	3.8663
legacy learnable interface rando	4	1500	0.2891	4.8512
legacy learnable interface fixed	0	0	0.0391	3.4783
legacy learnable interface fixed	0	150	1.0000	0.1207
legacy learnable interface fixed	0	1500	1.0000	0.0019
legacy learnable interface fixed	1	0	0.0215	3.4747
legacy learnable interface fixed	1	150	1.0000	0.0738
legacy learnable interface fixed	1	1500	1.0000	0.0017
legacy learnable interface fixed	2	0	0.0312	3.4700
legacy learnable interface fixed	2	150	1.0000	0.0581
legacy learnable interface fixed	2	1500	1.0000	0.0015
legacy learnable interface fixed	3	0	0.0449	3.4793
legacy learnable interface fixed	3	150	1.0000	0.1008
legacy learnable interface fixed	3	1500	1.0000	0.0027
legacy learnable interface fixed	4	0	0.0566	3.4705
legacy learnable interface fixed	4	150	1.0000	0.0731
legacy learnable interface fixed	4	1500	1.0000	0.0021
legacy trainable near-swap	0	0	0.0234	3.4641
legacy trainable near-swap	0	150	1.0000	0.0569
legacy trainable near-swap	0	1500	1.0000	0.0016
legacy trainable near-swap	1	0	0.0410	3.4781
legacy trainable near-swap	1	150	1.0000	0.0623
legacy trainable near-swap	1	1500	1.0000	0.0016
legacy trainable near-swap	2	0	0.0254	3.4697
legacy trainable near-swap	2	150	1.0000	0.0496
legacy trainable near-swap	2	1500	1.0000	0.0013
legacy trainable near-swap	3	0	0.0215	3.4606
legacy trainable near-swap	3	150	1.0000	0.1082
legacy trainable near-swap	3	1500	1.0000	0.0025
legacy trainable near-swap	4	0	0.0078	3.4713
legacy trainable near-swap	4	150	1.0000	0.0681
legacy trainable near-swap	4	1500	1.0000	0.0022

Table 33. All 65 per-seed curve values. The legacy random-interface rows stay between 0.21 and 0.29 at every step on every seed, far from this task's 1/32 chance and never above 0.30; the codebook-fixed rows reach 1.0 by step 150 on every seed and never regress. The analytic witness is exact at step 0. No seed was excluded and no value was rounded from a different run.

arm	seed	first full cover	committed	core bytes
fused gated	0	170	49	32768
fused gated	1	100	55	32768
fused gated	2	90	47	32768
fused gated	3	120	53	32768
fused gated	4	170	54	32768
no consolidation	0	none	0	32768
no consolidation	1	none	0	32768
no consolidation	2	none	0	32768
no consolidation	3	none	0	32768
no consolidation	4	none	0	32768
ungated clean	0	170	320	32768
ungated clean	1	100	320	32768
ungated clean	2	90	320	32768
ungated clean	3	120	320	32768
ungated clean	4	170	320	32768
ungated corrupt	0	none	320	32768
ungated corrupt	1	none	320	32768
ungated corrupt	2	none	320	32768
ungated corrupt	3	none	320	32768
ungated corrupt	4	none	320	32768
random write	0	none	154	32768
random write	1	none	153	32768
random write	2	none	154	32768
random write	3	none	155	32768
random write	4	none	155	32768
break alignment	0	none	49	32768
break alignment	1	none	55	32768
break alignment	2	none	47	32768
break alignment	3	none	53	32768
break alignment	4	none	54	32768
rho fast	0	none	49	32768
rho fast	1	none	55	32768
rho fast	2	none	47	32768
rho fast	3	none	53	32768
rho fast	4	none	54	32768

Table 34. All 35 fusion runs with their first full-cover round, number of committed experiences and core state size. The core state is constant across every arm, every seed and every round; only the bounded slot metadata changes. These are the individual values behind Table 16 and Table 18.

End of artifact. No further numeric claims appear anywhere in the submission.